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Prevalence of low back pain and associated factors among medical students in Wachemo University Southern Ethiopia

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Low back pain (LBP) is the most common musculoskeletal disorder in adults. According to previous studies medical students were found vulnerable for the development of LBP. We assessed the prevalence of LBP and associated factors among Medical students in Wachemo University, Southern Ethiopia. An institution-based cross-sectional study design and simple random sampling technique was employed to select 370 medical students. A structured pretested interviewer administered questionnaire was used to collect the data. The data were entered into Epi-info version 3.1 and analysed by SPSS version 25. Analytical and descriptive statistics were conducted. Simple and multiple logistic regression analyses were employed to assess factors associated with LBP. Variables at $p\text{-value} \leq 0.25$ in simple logistic regression analysis were entered into multiple logistic regression. Statistical significance was declared at $P\text{-value}, < 0.05$ in multiple logistic analysis. The prevalence of low back pain among medical students was found to be 40.1% with 95% CI (38.5–41.8). Being male students [AOR (95% CI): 1.83 (1.09–3.08)], design of study sitting chairs without back support [adjusted odds ratio (AOR) (95% C.I): 2.81 (1.74–4.55)], stress during class [AOR (95% C.I): 2.60 (1.56–4.35)], and daily study hours greater than 6 h and above [AOR: 4.33, 95% CI (2.30–8.12)] had a significant association with LBP. The prevalence of LBP in this study is high. Better ergonomic facilities and psychosocial supports to medical students are recommended to reduce the effects of predisposing factors on LBP.

Keywords Low back pain, Medical students, Musculoskeletal disorder, Southern Ethiopia

Lower back pain (LBP) is defined as pain and discomfort below the costal margin and above the inferior gluteal folds, with or without referred leg pain¹. It is the most common musculoskeletal disorders (MSD) in adults globally. The pain is felt by subjects from the back below the twelfth rib and above the inferior gluteal fold. Nerve roots, muscles, fasciae, bones, joints, intervertebral discs, and organs within the abdominal cavity are all potential anatomical origin of symptoms². LBP is neither a disease nor a diagnosable condition. It refers to pain of varying lengths in a part of the anatomy that has become a response to how people react to external and internal stimuli³.

LBP has multifactorial risk factors. According to previous studies; age, sex, body mass index, smoking, drinking habit, physical exercise, stress, sleeping hours, computer uses, long seating hours, seating without back support, duration of reading were found a risk factors for LBP^{4,5}.

LBP is the most frequent orthopaedic health problem that affects people of all ages around the world. It is the most common reason for seeking medical advice. It causes disability, affecting people's quality of life and occupational performance. Moreover, it has a significant financial impact on many countries' healthcare systems. Total cost associated with this illness is estimated to reach \$100 billion per year in the United States⁶. It is the leading cause of activity limitation and work absence throughout much of the world, and it causes an enormous economic burden on individuals, families and communities⁷.

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According to a systematic review and meta-analysis conducted in Africa, the reported annual prevalence of LBP in Africa was estimated to be 57%. It was higher than the global estimate of LBP. The reason might be priority is given to more critical health issues, LBP and other musculoskeletal diseases are less emphasized in lower and middle income countries (LMICs) such as Africa. This is most likely because, it is not life-threatening and health expenditures are already constrained and diverted to other, higher-priority illnesses. On the African continent, the prevalence of LBP is also increasing, but the limited information regarding this in developing countries is a major limitation^{5,8}.

A study conducted in Ethiopia at Gondar University revealed that the prevalence of MSD in any part of the body in the previous 12 months among medicine and nursing students was 69.4%. LBP was the most commonly reported MSD, and its prevalence increased among medical students as the year of study increased, ranging from 38% among 2nd year to 74.4% among 5th-year students⁹.

According to previous studies medical students were found vulnerable for the development of LBP^{10,11}. A study conducted at Malaysian Medical College, 46.1% of students had LBP in the previous year¹². A similar study done in Central Saudi Arabia LBP was reported to be 67% and 45.3% in different universities.^{13,14}

The nature of medical school and the medical curriculum has been associated with increased risks of LBP. The ward rotation and reading hours of medical students are sedentary most of the time, and the alignment of their chairs, tables, and computers is not designed based on their health aspects. In addition to sedentary life, poorly designed ergonomic chairs and tables lead to LBP, which in turn causes decreased productivity, missed class, and emotional stress, leading to depression⁵. There was still a scarcity of information about this vulnerable group in developing countries, including Ethiopia.

Although many studies conducted in different parts of Ethiopia regarding LBP, they were focused on teachers, hospital nurse staff, bank workers, office workers and car drivers^{15–19}. But there was limited information on the prevalence and associated factors of LBP among medical students across the country in general and in this study area in particular. Knowledge of the risk factors for LBP is important to prevent or reduce the incidence of LBP and related complications. Hence, the purpose of this study was to assess the prevalence and identify factors associated with LBP among medical students.

Methods and materials

Study setting, design and population

An Institution-based cross-sectional study design was conducted among medical students in Wachemo University from June to July 2022. Wachemo University is found at Hosanna city, Southern Ethiopia, located 230 km southwest from capital city of Ethiopia, Addis Ababa. It is the one among the growing government-funded universities in Ethiopia. The university has seven colleges and two campuses. There are 417 undergraduate medical students (male = 272, female = 145). Randomly selected undergraduate Medical students who fulfilled the inclusion criteria were participated in the study. Students with a well-known health problem: trauma or surgery around the low back area, using any supports or braces for the back, legs, or neck, females on premenstruation or menstruation period during data collection, history of disc prolapse before admission to campus and structural asymmetry like scoliosis or lordosis and kyphosis were excluded.

Sample Size determination and sampling procedure

Since the sample size calculated for second objective for assessing factors associated with LBP was greater than for first objective for determining the prevalence of LBP, the required sample size was calculated using double proportion formula using online epi-info considering the following assumptions.

Selected variable: Stress during classes, CI = 95%, Power = 80%, Percent of outcome in non-exposed = 13.1, OR = 2.3²⁰. By adding 10% of non-response rate the sample size was found to be 370²¹.

The study participants were categorized in to five strata based on academic years. After obtaining the list of students by their ID numbers, we selected the study participants using simple random sampling from each strata based on proportional allocation to sample size. (Fig. 1).

Study variables

Dependent variable

Low back pain

Independent variables

Socio-demographic, anthropometric, physical and ergonomic factors, behavioural and psycho-social factors.

Operational definition

Low back pain: defined as the experience pain or discomfort at least one episode in the past 12 months not related to trauma or menstrual pain²².

Lifting heavy objects: It refers to lifting, carrying, pulling, or pushing heavy loads weighing 20 kg and above every day or every other day².

Body mass index: weight in kilograms divided by the square of the height in meters (kg/m²). Underweight = BMI < 18.50, Normal range = BMI b/n 18.50– 24.99, Overweight = BMI b/n 25.00–29.99, and Obese = BMI > 30.00²².

Cigarette smoking: It is a practice of smoking cigarette by students for at least one sticks of cigarette per day²².

Alcohol drinking: it is a consumption of any kind of alcohol by students at least for two times per week²².

Physical exercise: Performing any kinds of physical exercise at least two times per week and for 30 min²².

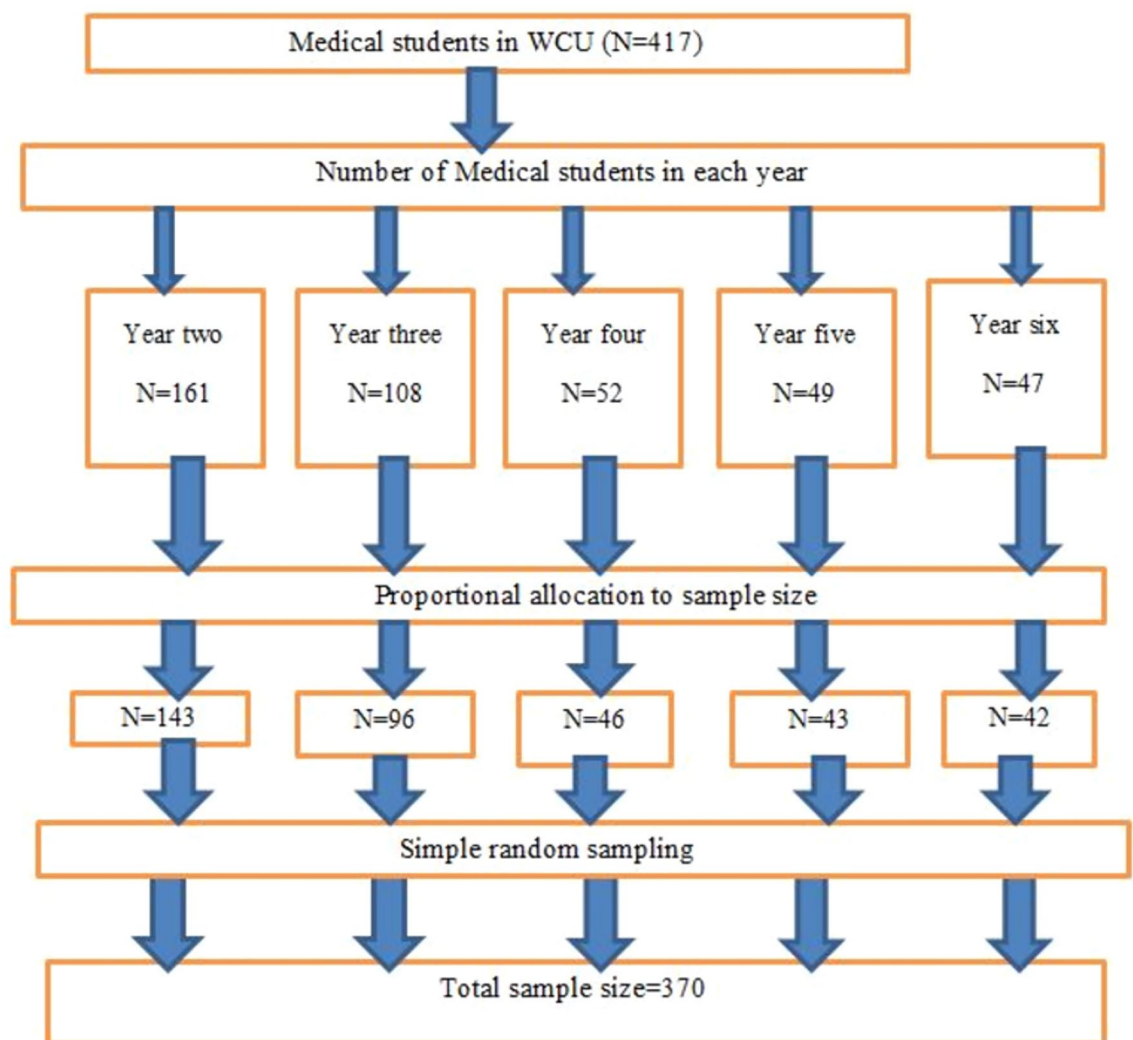


Fig. 1. Schematic representation of sampling procedures to assess the prevalence of low back pain and associated factors among Medical students in Wachemo University, Southern Ethiopia, 2022.

Perceived stress: Perceived Stress Scale (PSS) is a classic stress assessment instrument of medical students which range from 0 to 40²⁴. A score measured using the stress scale as yes^{24to40} was considered as having perceived stress and no (lower than or equal 23) was considered as not having perceived stress²².

Disability status: defined as assessment of disability status imposed by low back pain and categorized as score 0—4 (No disability), 5—14 (Mild disability), 15—24 (Moderate disability), 25—34 (Severe disability) and 35—50 (Completely disabled)¹⁴.

Incorrect sleep position: prone and flexed side sleep without pillow support¹⁴.

Ergonomic risk factors: are referring to workplace situations that cause wear and tear on the body and can cause injury²³.

Data collection tools and procedure

Data were collected using interviewer administered questionnaire and check list adapted from literatures. The questionnaire contained: Socio-demographic factors, behavioural and psycho-social factors, physical and ergonomic factors.

Anthropometric measurement for the height and weight of the participants was measured to calculate BMI. Five data collectors who were BSc degree holder nurses and two supervisors with MSc degree holders were participated in data collection and supervision respectively.

Weight was measured using a floor weighing scale with participants standing without shoes and wearing light clothing and recorded to the nearest 0.1 kg. Height was measured using a stadiometer while standing upright with the head in the Frankfort plane (a standard craniometrical reference plane passing through the right and left pterion and the left orbital) and recorded with an approximation of 1 cm.

The Oswestry Low Back Pain Disability Questionnaire was used to assess disability status due to low back pain. Accordingly it was categorized as score 0—4 (No disability), 5—14 (Mild disability), 15—24 (Moderate

disability), 25–34 (Severe disability) and 35–50 (Completely disabled). In addition, Perceived Stress Scale (PSS)-10 questionnaire was utilized to assess self-perceived stress.

Data processing and analysis

The data were entered into Epi-info version 3.1 and exported to SPSS version 25 for analysis. Descriptive statistics were computed using frequencies with percentages for categorical variables and mean and standard deviations for continuous variables. Both simple and multiple logistic regressions analysis were employed to assess factors associated with LBP.

Variables with a $P < 0.25$ in the simple logistic regression analysis were taken in to multiple logistic analyses and variables with $P < 0.05$ in the multiple logistic regression analysis were considered statically significant with adjusted odds ratio (AOR) with 95% CI. Model fitness was evaluated by Hosmer–Lemeshow test ($p = 0.649$). In the final model, multi-collinearity was measured using the variance inflation factor (VIF), which was found to be within a tolerable range (less than 10).

Data quality assurance

The questionnaires were pretested on 5% of the sample size outside the study area to assure its validity. Three days training was given for data collectors and supervisors on the purpose of study, skills of interview, data collection tools and ethical procedures. The supervisors were made onsite supervision during the data collection period and review all filled questionnaires so as to identify incomplete and incoherent responses. Data clean-up and cross-checking of missing data were done before analysis on SPSS.

To assess the reliability of the questionnaire, cronbach alpha was calculated for disability status and perceived stress assessing questionnaire and found to be 0.78 and 0.84, respectively. Besides, the validity of questionnaire was also assessed using Pearson correlation coefficient.

To minimize inter observer bias during anthropometric measurement, each data collectors measured the height and weight of individuals three times and they took the average value of the three measurements.

Results

Socio demographic and educational characteristics of the participants

From the total of 370 respondents, 367 were participated, making the response rate 99.2%. Three questionnaires were excluded for incomplete response. The mean age of the students was 23.4 ± 2.49 (SD) years with an age range between 19–30 years, and more than half of the participants 213 (58%) were in the age groups 22 and above. Nearly two thirds of the students 237(64.6%) were male, about 237(64.6%) were preclinical years, and more than one third of the students 142 (38.7%) were in second year (Table 1).

Behavioral and psycho-social characteristics of the participants

The majority of the participants 326(88.8%) had normal BMI ranging between 18.5–24.9 kg/m². About, 355(96.7%) had no physical exercise, almost all of the students 361(98.4%) did not have the habit of smoking. Similarly, most of the students, 337(94.6%) and 303(82.6%) did not have the habit of drinking alcohol and lifting/moving heavy objects, respectively.

Regarding psychosocial characteristics, nearly one third of the participants 111(30.2%) had perceived stress during the class. (Table 2).

Physical and ergonomic factors of the respondents

More than three-fourths 310(84.5%) of the respondents reported that the duration of computer/tablet use per day (hours) was 6 h and above. More than half of the students, 193(52.6%), mostly used both study table and bed as their usual place of study. The majority of students 298(81.2%) were reading more than 6 h per day, and nearly three-fourth of participants, 272(74.1%) were reading for 2–4 h continuously without a break. Regarding the sitting posture, more than four-fifths of the students, 305(83.1%), were frequently change their sitting positions

Variable	Categories	Frequency	Percentage (%)
Age	<22 years	154	42
	22 and above	213	58
Sex	Male	237	64.6
	Female	130	35.4
Academic year	Year two	142	38.7
	Year three	95	25.9
	Year four	45	12.3
	Year five	43	11.7
	Year six	42	11.4
Year of study	Preclinical year	237	64.6
	Clinical year	130	35.4

Table 1. Socio-demographic characteristics of medical students in Wachamo University, Southern Ethiopia.

Variables	Categories	Frequency	Percentage (%)
BMI	< 18.5	38	10.4
	18.5–24.9	326	88.8
	25–29.9	3	0.8
	≥ 30	0	0
Smoking cigarette	Yes	6	1.6
	No	361	98.4
Drinking alcohol	Yes	20	5.4
	No	337	94.6
Habit of exercise	Yes	12	3.3
	No	355	96.7
Lifting heavy object	Yes	64	17.4
	No	303	82.6
Perceived stress during class	Yes	111	30.2
	No	256	69.8

Table 2. Behavioral and psycho-social factors among medical students in Wachamo University, Southern Ethiopia.

during study. Among the study participants, more than half of respondents 194 (52.9%) mostly used a design of chair with back support, and more than three-quarters, 314(85.6%), slept more than 6 h per day (Table 3).

Prevalence of low back pain and its characteristics

In this study the prevalence of LBP was found to be 40.1% with 95% CI (38.5–41.8). Regarding the disability status, 10(6.8%) of individuals were severely disabled due to LBP. Nearly half of them, 68(46.3%), believed that sitting-prolonged period was considered as the possible cause of LBP.

Of the medical students who suffered from LBP, more than seven-eighths of participants, 128(87.1), reported that their first episode of LBP was observed after being admitted to campus. Out of the participants who suffered from LBP, class absenteeism was reported in 17 (11.6%) of students. Concerning aggravating factors, more than two fifth of respondents, 62(42.2%), reported that LBP worsened during prolonged sitting. (Table 4).

Factors associated with low back pain

Sex, academic year, design of study sitting chair, frequently of using computer/tablet, daily study hours/day, sitting posture during study, and perceived stress during class were selected for multiple logistic regression analysis at p -value < 0.25.

Variables	Categories	Frequency	Percentage (%)
Frequently use the computer/tablet	Yes	337	91.8
	No	30	8.2
Duration of computer/tablet use per day (hours)	< 6 h	57	15.5
	≥ 6 h	310	84.5
Usual place of study	Study table	145	39.5
	Bed	29	7.9
	Both	193	52.6
Daily study hours/ day	> = 6 h	69	18.8
	< 6 h	298	81.2
Study hours per day without taking rest break	2–4 h	272	74.1
	5–6 h	42	11.4
	> 6 h	53	14.4
Currently study sitting posture	Sitting Constant position for > 2 h	19	5.2
	Change sitting position per 2 h	43	11.7
	Frequently change sitting position	305	83.1
Design of study sitting chairs mostly	Have no back support	173	47.1
	Have back support	194	52.9
Average sleeping hours per day	≤ 6 h	53	14.4
	> 6 h	314	85.6

Table 3. Physical and Ergonomic factors among medical students in Wachamo University, Southern Ethiopia.

Variables	Categories	Frequency	Percentage (%)
Presence of LBP(n = 147)	Yes	147	40.1
	No	220	59.9
Disability status (147)	Not disability	47	31.9
	Mild	50	34
	Moderate	40	27.2
	Severe	10	6.8
Days it lasts this LBP(n = 147)	1 to 7 days	77	52.4
	8 to 30 days	31	21.1
	> 30 days	39	26.5
When you developed the pain(n = 147)	Before admission to campus	19	12.9
	After admission to campus	128	87.1
Nature of LBP(n = 147)	Gradually	89	60.5
	Suddenly	58	39.5
Cause LBP(n = 147)	Sitting prolonged	68	46.3
	Walking	9	6.1
	Standing-prolonged	41	27.89
	Lifting/move heavy object	29	19.72
Factors that increase the LBP(n = 147)	Exercises	3	2
	Carrying laptop bag	33	22.4
	Wrong sleep position	7	4.8
	Prolonged standing	29	19.7
	Prolonged sitting	62	42.2
	Lifting/move heavy object	13	8.8
Factors that relieves LBP(n = 147)	Sit in the right position	52	35.4
	Sleeping in the right position	53	36.1
	Massage	15	10.2
	Medication	7	4.8
	Exercise	20	13.6
Frequency of LBP(n = 147)	Only once	8	5.4
	Sometimes	126	85.7
	Frequently	13	8.8
Got health care for LBP(n = 147)	Yes	22	14.9
	No	125	85.0
Missed class because of pain(n = 147)	Yes	17	11.6
	No	130	88.4

Table 4. Prevalence and characteristics of low back pain among medical students in Wachamo University, Southern Ethiopia.

In multiple logistic analysis with the hierarchical entry method study sitting chairs without back support, being male students, perceived stress during class and daily study hours per day greater than 6 h and above were significantly associated with LBP at $p < 0.05$.

Being male students were 1.83 times more likely to develop low back pain compared to female students [AOR = 1.83, 95% CI: (1.09–3.08)]. The odds of developing LBP among students who had perceived stress during class were 2.6 times more likely to develop LBP compared to those who had no perceived stress [AOR = 2.60, 95% CI: (1.56–4.35)]. There was also a statistical association between daily study hours per day and LBP among students. Students who read greater than 6 h and above per day were 4.33 times more likely to develop LBP compared to students who read less than 6 h per day [AOR = 4.33, 95%CI: (2.30–8.12)].

Students who sat on chairs that do not have back support during reading had 2.81 times higher odds of having LBP compared to those who used chairs that had back support [AOR = 2.81, 95%CI: (1.74–4.55)] (Table 5).

Discussion

This is the first study in its kind for the study area in particular and in Ethiopia in general to assess the prevalence of LBP and its associated factors by including the medical students as study participants. In this study, the prevalence of LBP among medical students was found to be 40.1% with 95%CI (38.5–41.8), which was in line with study done in China (40.1%)²⁴.

However, the finding of the present study was lower than studies conducted in Saudi Arabia (67.6%)¹⁰, Bangladesh (63.3%)²⁵, Serbia (59.5%)²⁶, France (72%)²⁷, Gondar (55.6%)⁹ and Egypt (73.1%)¹¹. This variation might be due

Variables	Category	LBP		COR with 95%CI	AOR with 95% CI	P-value
		Yes	No			
Sex	Male	110	127	2.18(1.4–3.44)	1.83(1.09–3.08)	0.02*
	Female	37	93	1	1	
Academic year	Year two	55	87	1	1	
	Year three	46	49	0.67(0.4–1.14)	0.71(0.40–1.28)	0.26
	Year four	12	33	1.74(0.83–3.65)	2.06(0.90–4.70)	0.09
	Year five	21	22	0.66(0.33–1.32)	0.68(0.31–1.49)	0.34
	Year six	13	29	1.41(0.68–2.95)	2.24(0.96–5.20)	0.06
Design of study sitting chairs	No back support	87	86	2.26(1.48–3.46)	2.81(1.74–4.55)	0.000*
	Back support	60	134	1	1	
Frequently use computer	Yes	131	206	0.56(0.26–1.18)	0.90(0.39–2.13)	0.80
	No	16	14	1	1	
Sitting posture	Sitting Constant position for > 2 h	5	14	1	1	
	Change sitting position/hr	16	27	0.60(0.18–1.99)	0.48(0.13–1.79)	0.27
	Frequently change sitting position	126	179	0.51(0.18–1.44)	0.46(0.15–1.42)	0.18
Stress	Yes	62	49	2.55(1.61–4.02)	2.60(1.56–4.35)	0.000*
	No	85	171	1	1	
Daily study-hour/day	> = 6 h	45	24	3.6(2.08–6.25)	4.33(2.30–8.12)	0.000*
	< 6 h	102	196	1	1	

Table 5. Simple and multiple logistic regression analysis for factors associated with LBP among medical students in Wachamo University, Southern Ethiopia. *significant at P-value < 0.05.

to differences in the methodology, definition of LBP used in each study, sample size, awareness and openness to the questions, individual perception of pain and differences in assessment tools.

On another hand, it was higher than studies conducted in Pakistan(21.2%)²⁶ and Saudi Arabia (30%)⁴. The possible reason might be due to poor health facilities, socioeconomic differences and low awareness about capabilities on handling school ergonomics.

In this study, being male students, perceived stress during class, study sitting chairs without back support, and study hours per day greater than 6 h and above were factors significantly associated with LBP.

The odds of developing LBP among male students were 1.83 more likely than female students. In contrast to current study, studies conducted in Saudi Arabia²⁸ and Serbia²⁶ showed that being female students were significantly associated with occurrence of LBP compared to males. This might be due to conditions specific to females like premenstrual syndrome, premenstrual dysmorphic disorder, menstrual cycle fluctuations, and dysmenorrhea being excluded in this study in contrast to the above studies.

In this study, stress during class was significantly associated with LBP. The odds of LBP were 2.6 times higher among medical students who had perceived stress during class as compared with those not having perceived stress. This finding was supported by studies conducted in Serbia²⁶ and Saudi Arabia²⁹. This might be caused by mental tension or stress, which reduces blood supply to the body's soft tissues, including the muscles, tendons, ligaments, and nerves in the back. This will result in a further reduction in oxygen delivery to all tissues as well as a build-up of waste products in the muscles. The individual will consequently experience muscle tension, spasms, and back pain⁴. Also, high stress levels have been associated with the stimulation of the sympathetic nervous system, prompting the release of stress mediators that can strain the musculoskeletal system, resulting in LBP.

In this study, studying hours per day were significantly associated with low back pain. Students who sat for greater than 6 h and above per day for study were 4.33 times more likely to have low back pain compared to those who sat for less than 6 h per day. This finding was consistent with studies conducted in Bangladesh²⁵, Saudi Arabia⁶ and Egypt¹¹. The possible reason might be that continued reading without a break results in uncomfortable postures that contribute to back problems and muscle strains. It causes the Para spinal muscles to malfunction and adds to the load on the spine. This is due to the fact that prolonged sitting and restricted spinal motion may increase back stress, placing additional strain on the back's muscles and supporting tissues and causing joint pain, spinal disc compression, and muscle tiredness. Lack of spinal mobility causes the spinal ligaments to stretch and hinders the passage of fluid into the vertebral disc, which ultimately results in the vertebral discs receiving insufficient nutrition. This prolonged activity leads to impaired oxygenation of muscle tissues and has been implicated as a cause of back pain⁴.

This study showed that students who sat in chairs that do not have back support during reading had 2.81 times higher odds of having LBP compared to those who used chairs that had back support. This finding is consistent with the studies conducted in Bangladesh²⁵ and Saudi Arabia⁶. This might be caused by sitting on a chair that does not have back support during reading, which forces the muscular system to support body structure, which leads to the spine being bent out of its natural shape, which in turn has negative effects on and damage to the muscular system over time. In addition, it may impair the ability of the postural muscles to support the body and also lead to abnormal strain of the neuromuscular system, consequently causing pain.

In this study, academic year didn't show any significant association with the occurrence of low back pain. In contrast to this study, studies conducted in France²⁷ and Saudi Arabia²⁸ showed that being in 3rd year and 2nd year medical studies had a significant association with the occurrence of low back pain, respectively. Possible reason might be differences in faculty year of study, burden of academic curriculum, methodological heterogeneity, mode of data collection, and the definition of back pain used in the study.

Strengths and limitations of the study

As strength it is the first study in its kind by including the medical students as study participants which are vulnerable groups for developing LBP. However the study was not without limitations. Firstly as it is a cross sectional study design causality between dependent and independent variables couldn't be determined. Secondly, since the study was conducted in one facility, the results may not be representative of all medical students in Ethiopia.

Conclusions and recommendations

The prevalence of LBP among medical students in this study was 40.1%, higher than previous studies. The study found that sex, study sitting chairs without back support, daily study hours per day greater than 6 h and above, and stress during class were significantly associated with LBP. Better ergonomic facilities and psychosocial support to medical students were recommended to reduce the effects of predisposing factors on LBP.

Data availability

The datasets used and/or analyzed during the current study available from the corresponding author on reasonable request.

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Author contributions

TG and MF involved in conceptualization, study design, execution, data gathering, analysis, and interpretation. AG, MB and MS participated to the article's drafting, revision, or critical evaluation. All authors gave the document final approval before submission to the journal.

Competing interests

The authors declare no competing interests.

Ethical approval

Ethical approval was obtained from Arba Minch University, Institutional research Ethics Review Board, College of Medicine and Health Sciences with reference number (IRB/1253/2022). Informed written consent was taken from all involved participants. The aim of the study was explained for each study participants. Participants were also informed that they could withdraw from the study at any time if they were not comfortable about the questions. To ensure confidentiality, their names, and other personal identifiers were not registered in the survey tool. Moreover, this study was conducted in accordance with the Declaration of Helsinki, and all ethical and professional considerations were maintained throughout the study to keep participants' data strictly confidential. Students who had experienced persistent pain were linked to the student Clinic at WCU for further investigation and treatment.

Additional information

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